

隐函数的求导法则

一、一个方程的情形

1. $F(x, y) = 0$

隐函数存在定理 1 设函数 $F(x, y)$ 在点 $P(x_0, y_0)$ 的某一邻域内具有连续的偏导数, 且 $F(x_0, y_0) = 0$, $F_y(x_0, y_0) \neq 0$, 则方程 $F(x, y) = 0$ 在点 $P(x_0, y_0)$ 的某一邻域内恒能唯一确定一个单值连续且具有连续导数的函数 $y = f(x)$, 它满足条件 $y_0 = f(x_0)$, 并

有
$$\frac{dy}{dx} = -\frac{F_x}{F_y}.$$

例 1 验证方程 $x^2 + y^2 - 1 = 0$ 在点 $(0,1)$ 的某邻域内能唯一确定一个单值可导、且 $x = 0$ 时 $y = 1$ 的隐函数 $y = f(x)$ ，并求这函数的一阶和二阶导数在 $x = 0$ 的值.

解 令 $F(x, y) = x^2 + y^2 - 1$

则 $F_x = 2x, \quad F_y = 2y,$

$$F(0,1) = 0, \quad F_y(0,1) = 2 \neq 0,$$

依定理知方程 $x^2 + y^2 - 1 = 0$ 在点 $(0,1)$ 的某邻域内能唯一确定一个单值可导、且 $x = 0$ 时 $y = 1$ 的函数 $y = f(x)$.

函数的一阶和二阶导数为

$$\frac{dy}{dx} = -\frac{F_x}{F_y} = -\frac{x}{y}, \quad \left. \frac{dy}{dx} \right|_{x=0} = 0,$$

$$\frac{d^2 y}{dx^2} = -\frac{y - xy'}{y^2} = -\frac{y - x\left(-\frac{x}{y}\right)}{y^2} = -\frac{1}{y^3},$$

$$\left. \frac{d^2 y}{dx^2} \right|_{x=0} = -1.$$

例 2 已知 $\ln \sqrt{x^2 + y^2} = \arctan \frac{y}{x}$, 求 $\frac{dy}{dx}$.

解 令 $F(x, y) = \ln \sqrt{x^2 + y^2} - \arctan \frac{y}{x},$

则 $F_x(x, y) = \frac{x + y}{x^2 + y^2}, \quad F_y(x, y) = \frac{y - x}{x^2 + y^2},$

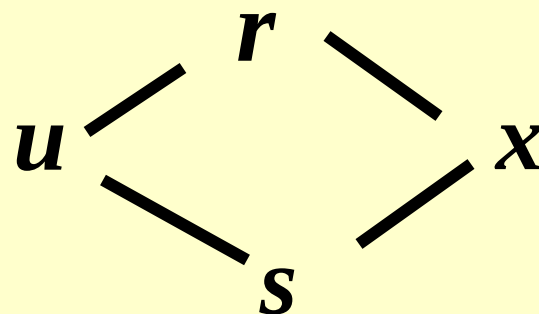
$$\frac{dy}{dx} = -\frac{F_x}{F_y} = -\frac{x + y}{y - x}.$$

2. $F(x, y, z) = 0$

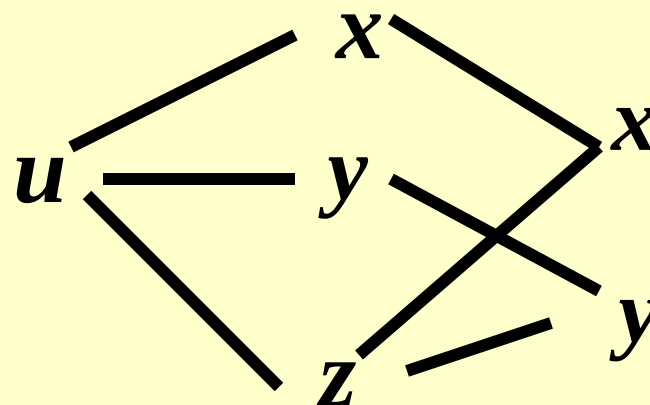
隐函数存在定理 2 设函数 $F(x, y, z)$ 在点 $P(x_0, y_0, z_0)$ 的某一邻域内有连续的偏导数, 且 $F(x_0, y_0, z_0) = 0$, $F_z(x_0, y_0, z_0) \neq 0$, 则方程 $F(x, y, z) = 0$ 在点 $P(x_0, y_0, z_0)$ 的某一邻域内恒能唯一确定一个单值连续且具有连续偏导数的函数 $z = f(x, y)$, 它满足条件 $z_0 = f(x_0, y_0)$,

并有 $\frac{\partial z}{\partial x} = -\frac{F_x}{F_z}, \quad \frac{\partial z}{\partial y} = -\frac{F_y}{F_z}.$

$$u = F(r, s), r = x, s = y(x)$$



$$u = F(x, y, z), z = z(x, y)$$



例 3 设 $x^2 + y^2 + z^2 - 4z = 0$, 求 $\frac{\partial^2 z}{\partial x^2}$.

解 令 $F(x, y, z) = x^2 + y^2 + z^2 - 4z$,

$$\text{则 } F_x = 2x, \quad F_z = 2z - 4, \quad \frac{\partial z}{\partial x} = -\frac{F_x}{F_z} = \frac{x}{2-z},$$

$$\begin{aligned} \frac{\partial^2 z}{\partial x^2} &= \frac{(2-z) + x \frac{\partial z}{\partial x}}{(2-z)^2} = \frac{(2-z) + x \cdot \frac{x}{2-z}}{(2-z)^2} \\ &= \frac{(2-z)^2 + x^2}{(2-z)^3}. \end{aligned}$$

例 4 设 $z = f(x + y + z, xyz)$, 求 $\frac{\partial z}{\partial x}, \frac{\partial x}{\partial y}, \frac{\partial y}{\partial z}$.

思路: 把 z 看成 x, y 的函数对 x 求偏导数得 $\frac{\partial z}{\partial x}$,

把 x 看成 z, y 的函数对 y 求偏导数得 $\frac{\partial x}{\partial y}$,

把 y 看成 x, z 的函数对 z 求偏导数得 $\frac{\partial y}{\partial z}$.

解 令 $u = x + y + z$, $v = xyz$,

则 $z = f(u, v)$,

把 z 看成 x, y 的函数对 x 求偏导数得

$$\frac{\partial z}{\partial x} = f_u \cdot (1 + \frac{\partial z}{\partial x}) + f_v \cdot (yz + xy \frac{\partial z}{\partial x}),$$

整理得
$$\frac{\partial z}{\partial x} = \frac{f_u + yzf_v}{1 - f_u - xyf_v},$$

把 x 看成 z, y 的函数对 y 求偏导数得

$$0 = f_u \cdot \left(\frac{\partial x}{\partial y} + 1 \right) + f_v \cdot (xz + yz \frac{\partial x}{\partial y}),$$

整理得 $\frac{\partial x}{\partial y} = -\frac{f_u + xzf_v}{f_u + yzf_v},$

把 y 看成 x, z 的函数对 z 求偏导数得

$$1 = f_u \cdot \left(\frac{\partial y}{\partial z} + 1 \right) + f_v \cdot (xy + xz \frac{\partial y}{\partial z}),$$

整理得 $\frac{\partial y}{\partial z} = \frac{1 - f_u - xyf_v}{f_u + xzf_v}.$

二、方程组的情形

1、对于方程组
$$\begin{cases} F(x,y,z)=0 \\ \Phi(x,y,z)=0 \end{cases}$$

怎样求偏导数

首先应明确这个方程组确定了几个几元隐函数

当 x 给定以后相当于解含关于 y, z 的方程组

如果有解且唯一则对于不同的 x 就完全确定了 y, z

故方程组确定了两个一元隐函数 $y=y(x), z=z(x)$

怎样求 $\frac{dy}{dx}, \frac{dz}{dx}$ $F(x, y, z) = 0$ 两边对 x 求导

注意左边是复合函数（三个中间变量），

$$F_x + F_y \cdot \frac{dy}{dx} + F_z \cdot \frac{dz}{dx} = 0 \quad \text{同理}$$

$$\Phi_x + \Phi_y \cdot \frac{dy}{dx} + \Phi_z \cdot \frac{dz}{dx} = 0$$

若 $J = \begin{vmatrix} F_y & F_z \\ \Phi_y & \Phi_z \end{vmatrix} \neq 0$ 则

$$\frac{dy}{dx} = -\frac{1}{J} \begin{vmatrix} F_x & F_z \\ \Phi_x & \Phi_z \end{vmatrix}, \quad \frac{dz}{dx} = -\frac{1}{J} \begin{vmatrix} F_y & F_x \\ \Phi_y & \Phi_x \end{vmatrix}$$

$$2、\begin{cases} F(x, y, u, v) = 0 \\ G(x, y, u, v) = 0 \end{cases}$$

隐函数存在定理 3 设 $F(x, y, u, v)$ 、 $G(x, y, u, v)$ 在点 $P(x_0, y_0, u_0, v_0)$ 的某一邻域内有对各个变量的连续偏导数，且 $F(x_0, y_0, u_0, v_0) = 0$ ， $G(x_0, y_0, u_0, v_0) = 0$ ，且偏导数所组成的函数行列式（或称雅可比式）

$$J = \frac{\partial(F, G)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial F}{\partial u} & \frac{\partial F}{\partial v} \\ \frac{\partial G}{\partial u} & \frac{\partial G}{\partial v} \end{vmatrix}$$

在点 $P(x_0, y_0, u_0, v_0)$ 不等于零, 则方程组

$$F(x, y, u, v) = 0, \quad G(x, y, u, v) = 0$$

在点 $P(x_0, y_0, u_0, v_0)$ 的某一邻域内恒能唯一确定一组单值连续且具有连续偏导数的函数 $u = u(x, y)$, $v = v(x, y)$, 它们满足条件 $u_0 = u(x_0, y_0)$, $v_0 = v(x_0, y_0)$, 并有

$$\frac{\partial u}{\partial x} = -\frac{1}{J} \frac{\partial(F, G)}{\partial(x, v)} = -\frac{\begin{vmatrix} F_x & F_v \\ G_x & G_v \end{vmatrix}}{\begin{vmatrix} F_u & F_v \\ G_u & G_v \end{vmatrix}},$$

$$\frac{\partial v}{\partial x} = -\frac{1}{J} \frac{\partial(F, G)}{\partial(u, x)} = -\frac{\begin{vmatrix} F_u & F_x \\ G_u & G_x \end{vmatrix}}{\begin{vmatrix} F_u & F_v \\ G_u & G_v \end{vmatrix}},$$

$$\frac{\partial u}{\partial y} = -\frac{1}{J} \frac{\partial(F, G)}{\partial(y, v)} = -\frac{\begin{vmatrix} F_y & F_v \\ G_y & G_v \end{vmatrix}}{\begin{vmatrix} F_u & F_v \\ G_u & G_v \end{vmatrix}},$$

$$\frac{\partial v}{\partial y} = -\frac{1}{J} \frac{\partial(F, G)}{\partial(u, y)} = -\frac{\begin{vmatrix} F_u & F_y \\ G_u & G_y \end{vmatrix}}{\begin{vmatrix} F_u & F_v \\ G_u & G_v \end{vmatrix}}.$$

例5 设 $xu - yv = 0$, $yu + xv = 1$,

求 $\frac{\partial u}{\partial x}$, $\frac{\partial u}{\partial y}$, $\frac{\partial v}{\partial x}$ 和 $\frac{\partial v}{\partial y}$.

解1 直接代入公式;

解2 运用公式推导的方法,

将所给方程的两边对 x 求导并移项

$$\begin{cases} x \frac{\partial u}{\partial x} - y \frac{\partial v}{\partial x} = -u \\ y \frac{\partial u}{\partial x} + x \frac{\partial v}{\partial x} = -v \end{cases}, \quad J = \begin{vmatrix} x & -y \\ y & x \end{vmatrix} = x^2 + y^2,$$

在 $J \neq 0$ 的条件下,

$$\frac{\partial u}{\partial x} = \frac{\begin{vmatrix} -u & -y \\ -v & x \end{vmatrix}}{\begin{vmatrix} x & -y \\ y & x \end{vmatrix}} = -\frac{xu + yv}{x^2 + y^2}, \quad \frac{\partial v}{\partial x} = \frac{\begin{vmatrix} x & -u \\ y & -v \end{vmatrix}}{\begin{vmatrix} x & -y \\ y & x \end{vmatrix}} = \frac{yu - xv}{x^2 + y^2},$$

将所给方程的两边对 y 求导，用同样方法得

$$\frac{\partial u}{\partial y} = \frac{xv - yu}{x^2 + y^2}, \quad \frac{\partial v}{\partial y} = -\frac{xu + yv}{x^2 + y^2}.$$

注

这组公式不太好记，具体做题时应用的是其基本思想

关于隐函数求二阶偏导数

以 $F(x, y, z) = 0$ 为例，主要有三种方法：

①公式法 $\frac{\partial z}{\partial x} = -\frac{F_x}{F_z},$

$$\frac{\partial^2 z}{\partial x^2} = -\frac{\frac{\partial}{\partial x}(F_x)F_z - F_x \frac{\partial}{\partial x}(F_z)}{F_z^2}$$

$$= -\frac{1}{F_z^3} [F_{xx}F_z^2 - 2F_{xz}F_xF_z + F_{zz}F_x^2]$$

类似地可求得 $\frac{\partial^2 z}{\partial x \partial y}, \frac{\partial^2 z}{\partial y^2}$

②直接法 方程两边连续求导两次

$$F_x + F_z \cdot \frac{\partial z}{\partial x} = 0$$

$$F_{xx} + 2F_{xz} \cdot \frac{\partial z}{\partial x} + F_{zz} \left(\frac{\partial z}{\partial x} \right)^2 + F_z \frac{\partial^2 z}{\partial x^2} = 0$$

$$\text{解得: } \frac{\partial^2 z}{\partial x^2} = -\frac{1}{F_z^3} [F_{xx} F_z^2 - 2F_{xz} F_x F_z + F_{zz} F_x^2]$$

两种方法相比，法二较简便，因为可避免商的求导运算，尤其是在求指定点的二阶偏导数时，毋须解出一阶偏导数而是将其具体数值代入即可求得二阶偏导数，使运算大为简化。

③全微分法

利用全微分形式不变性，在所给的方程两边直接求全微分

$$dz = A dx + B dy \quad \text{则} \quad A = \frac{\partial z}{\partial x}, B = \frac{\partial z}{\partial y}$$

这样一次就可求得全部的一阶偏导数。

三、小结

隐函数的求导法则 （分以下几种情况）

$$(1) \quad F(x, y) = 0$$

$$(2) \quad F(x, y, z) = 0$$

$$(3) \quad \begin{cases} F(x, y, z) = 0 \\ \Phi(x, y, z) = 0 \end{cases}$$

$$(4) \quad \begin{cases} F(x, y, u, v) = 0 \\ G(x, y, u, v) = 0 \end{cases}$$

思考题

已知 $\frac{x}{z} = \varphi\left(\frac{y}{z}\right)$, 其中 φ 为可微函数,

$$\text{求 } x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = ?$$

思考题解答

$$\text{记 } F(x, y, z) = \frac{x}{z} - \varphi\left(\frac{y}{z}\right),$$

$$F_y = -\varphi'\left(\frac{y}{z}\right) \cdot \frac{1}{z}, \quad F_z = \frac{-x}{z^2} - \varphi'\left(\frac{y}{z}\right) \cdot \frac{(-y)}{z^2},$$

$$\frac{\partial z}{\partial x} = -\frac{F_x}{F_z} = \frac{z}{x - y\varphi'\left(\frac{y}{z}\right)}, \quad \frac{\partial z}{\partial y} = -\frac{F_y}{F_z} = \frac{-z\varphi'\left(\frac{y}{z}\right)}{x - y\varphi'\left(\frac{y}{z}\right)},$$

$$\text{于是 } x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = z.$$

练习题

一、填空题:

1、设 $\ln \sqrt{x^2 + y^2} = \arctan \frac{y}{x}$, 则

$$\frac{dy}{dx} = \underline{\hspace{10cm}}.$$

2、设 $z^x = y^z$, 则

$$\frac{\partial z}{\partial x} = \underline{\hspace{10cm}},$$

$$\frac{\partial z}{\partial y} = \underline{\hspace{10cm}}.$$

二、设 $2\sin(x + 2y - 3z) = x + 2y - 3z$,

$$\text{证明: } \frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} = 1.$$

三、如果函数 $f(x, y, z)$ 对任何 t 恒满足关系式 $f(tx, ty, tz) = t^k f(x, y, z)$, 则称函数 $f(x, y, z)$ 为 k 次齐次函数, 试证: k 次齐次函数满足方程

$$x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} + z \frac{\partial f}{\partial z} = k f(x, y, z).$$

四、设 $z^3 - 3xyz = a^3$, 求 $\frac{\partial^2 z}{\partial x \partial y}$.

五、求由下列方程组所确定的函数的导数或偏导数:

1、设 $\begin{cases} z = x^2 + y^2 \\ x^2 + 2y^2 + 3z^2 = 20 \end{cases}$, 求 $\frac{dy}{dx}, \frac{dz}{dx}$.

2、设 $\begin{cases} u = f(ux, v + y) \\ v = g(u - x, v^2 y) \end{cases}$, 求 $\frac{\partial u}{\partial x}, \frac{\partial v}{\partial x}$.

(其中 f, g 具有一阶连续偏导数)

六、设函数 $u(x)$ 由方程组
$$\begin{cases} u = f(x, y) \\ g(x, y, z) = 0 \\ h(x, z) = 0 \end{cases}$$
 所确定,

且 $\frac{\partial g}{\partial y} \neq 0, \frac{\partial h}{\partial z} \neq 0$, 求 $\frac{du}{dx}$. (f, g, h 均可微)

七、设 $y = f(x, t)$, 而 t 是由方程 $F(x, y, t) = 0$ 所确定的 x, y 的函数, 求 $\frac{dy}{dx}$.

八、设 $z = z(x, y)$ 由方程 $F(x + \frac{x}{y}, y + \frac{z}{x}) = 0$ 所确定,

证明: $x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = z - xy$.

练习题答案

一、 1、 $\frac{x+y}{x-y}$; 2、 $\frac{-z^x \ln z}{xz^{x-1} - y^z \ln y}$;

3、 $\frac{zy^{z-1}}{xz^{x-1} - y^z \ln y}$.

四、 $\frac{\partial^2 z}{\partial x \partial y} = \frac{z(z^4 - 2xyz^2 - x^2 y^2)}{(z^2 - xy)^3}$.

五、 1、 $\frac{dy}{dx} = \frac{-x(6z+1)}{2y(3z+1)}, \frac{dz}{dx} = \frac{x}{3z+1}$;

2、 $\frac{\partial u}{\partial x} = \frac{-uf'_1(2yvg'_2 - 1) - f'_2 \cdot g'_1}{(xf'_1 - 1)(2yvg'_2 - 1) - f'_2 \cdot g'_1},$

$$\frac{\partial v}{\partial x} = \frac{g'_1(xf'_1 + uf'_1 - 1)}{(xf'_1 - 1)(2yvg'_2 - 1) - f'_2 \cdot g'_1}.$$

$$\begin{aligned}
 \text{六、} \quad \frac{du}{dx} &= f'_x - \frac{f'_x \cdot g'_x}{g'_y} + \frac{f'_y \cdot g'_z \cdot h'_x}{g'_y \cdot h'_z} \\
 &= \frac{f'_x g'_y h'_z - f'_x g'_x h'_z + f'_y g'_z h'_x}{g'_y h'_z}.
 \end{aligned}$$

$$\text{七、} \quad \frac{dy}{dx} = \frac{F'_t \cdot f'_x - F'_x \cdot f'_t}{F'_t + F'_y \cdot f'_t}.$$